

RITWIK BISWAS

Software Engineer | Backend, Distributed Systems & AI

Kolkata, India | thisisritwikbiswas@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Indian Institute of Technology Kharagpur | 2021 - 2026

Dual Degree | B.Tech + M.Tech | Mechanical Engineering | CGPA: 7.84 / 10

Shyamal Ghosh & Sunanda Ghosh Endowment Honour

EXPERIENCE

Taskly Technologies Inc. | Software Engineering Intern

Remote, Canada | Mar 2025 - Jul 2025

- Built core JobSense modules with Next.js, FastAPI, and PostgreSQL for job-description analysis, resume scoring, ATS checks, interview preparation, and company research.
- Integrated Groq, Firebase Authentication, JWT-secured sessions, and protected AI endpoints; resolved scraping, payload, environment, and Docker failures while supporting Vercel and cloud deployment workflows.

Search-in | Full-Stack Developer Intern

Remote, Bengaluru | Jun 2024 - Jul 2024

- Built a React administration platform and Node.js/Express APIs supporting operational analytics, order workflows, 10K+ users, 1K+ orders, and 500+ managed records.
- Improved data paths with MongoDB indexing, Redis caching, and query tuning; redesigned workflows that reduced order-processing time by 30% and deployed frontend assets to AWS S3.

Pepcorns | Full-Stack Developer Intern

Remote, Pune | Mar 2024 - Apr 2024

- Built a Node.js/SQL referral and rewards platform with referral attribution, reward issuance, social sharing, backend integrations, and reusable React components.

FLAGSHIP ENGINEERING SYSTEMS

ThesisLens | Applied AI / Retrieval / Evaluation

Python | FastAPI | Azure OpenAI | Azure AI Search | PostgreSQL | Docker

- Built evidence-grounded financial research foundations spanning filing ingestion, evidence-guarded KPI extraction, deterministic SQL/RAG routing, and current-request citation validation.
- Evaluated 44 queries and 1,012 judged query-document pairs. On an eight-query frozen holdout, a fine-tuned MiniLM reranker moved nDCG@5 from 0.704 to 0.811 and Recall@3 from 0.448 to 0.604; lexical retrieval remained the serving default after a top-rank MRR tradeoff and approximately 3.19 s/query local CPU reranking latency.

TraceForge | AI Reliability / Deterministic Replay

Python | Go | Kafka | SQLite | LangGraph | OpenTelemetry | Docker

- Sealed sanitized execution evidence into versioned Replay Capsules and replayed recorded model and HTTP boundaries offline using ordered request fingerprints that fail closed on mismatch.
- Added normalized structural comparison, developer-authored regression specifications, and pytest export; the optional capture path uses a Go gateway, Kafka at-least-once delivery, and SQLite event-ID deduplication.

Converge | Distributed Real-Time Collaboration

TypeScript | React | Konva | Fastify | Socket.IO | PostgreSQL | Redis | IndexedDB

- Built board-authoritative state with a strict monotonic total order per board and atomic PostgreSQL operation, projection, receipt, heads, undo-evidence, and outbox updates; Redis Streams provide retained at-least-once fanout with application-level idempotency.
- Implemented persistence-before-optimism, fixed-watermark catch-up, generation-fenced reconnect, and verified snapshot-plus-tail recovery. Release evidence records 245 PostgreSQL integration tests across 29 files, 59 failure scenarios, 93 Playwright scenarios, and bounded local k6 runs.

TECHNICAL SKILLS

Languages: TypeScript, Go, Python, SQL, JavaScript

Systems: PostgreSQL, Redis, Kafka, Fastify, FastAPI, Socket.IO, IndexedDB

AI / Retrieval: LangGraph, Azure OpenAI, Azure AI Search, RAG, BM25, SentenceTransformers, FAISS

Quality / Infrastructure: Docker, GitHub Actions, Playwright, k6, OpenTelemetry, Prometheus

ACHIEVEMENTS

- LeetCode:** Knight; peak 1923; Top 5.6%; 1,550+ solved; 600+ Problem-of-the-Day streak.
- Codeforces:** Specialist; peak 1415; Global Rank 818 in Round 952.
- Amazon ML Summer School 2024; Top 3 in GCOS 2024 at IIT Kharagpur; Top 5 in Overnite at Kshitij, IIT Kharagpur.